MOTIVATION

Teaching traffic lights to cooperate

Multi-agent reinforcement learning on a 2<2 SUMO grid, aimed at one number: average waiting time.

Fixed timers

Cycle on a schedule and ignore live queues, so cars wait at empty junctions.

Linked junctions

One signal's choice changes the traffic its neighbours receive.

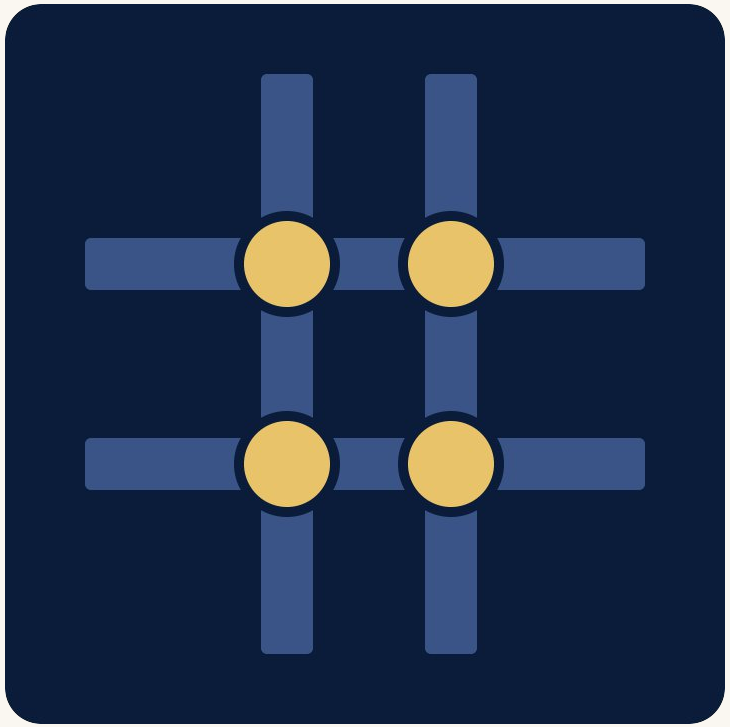
Our goal

Learn coordinated control that beats fixed timing on waiting time.

METHOD

One agent per junction, one shared goal

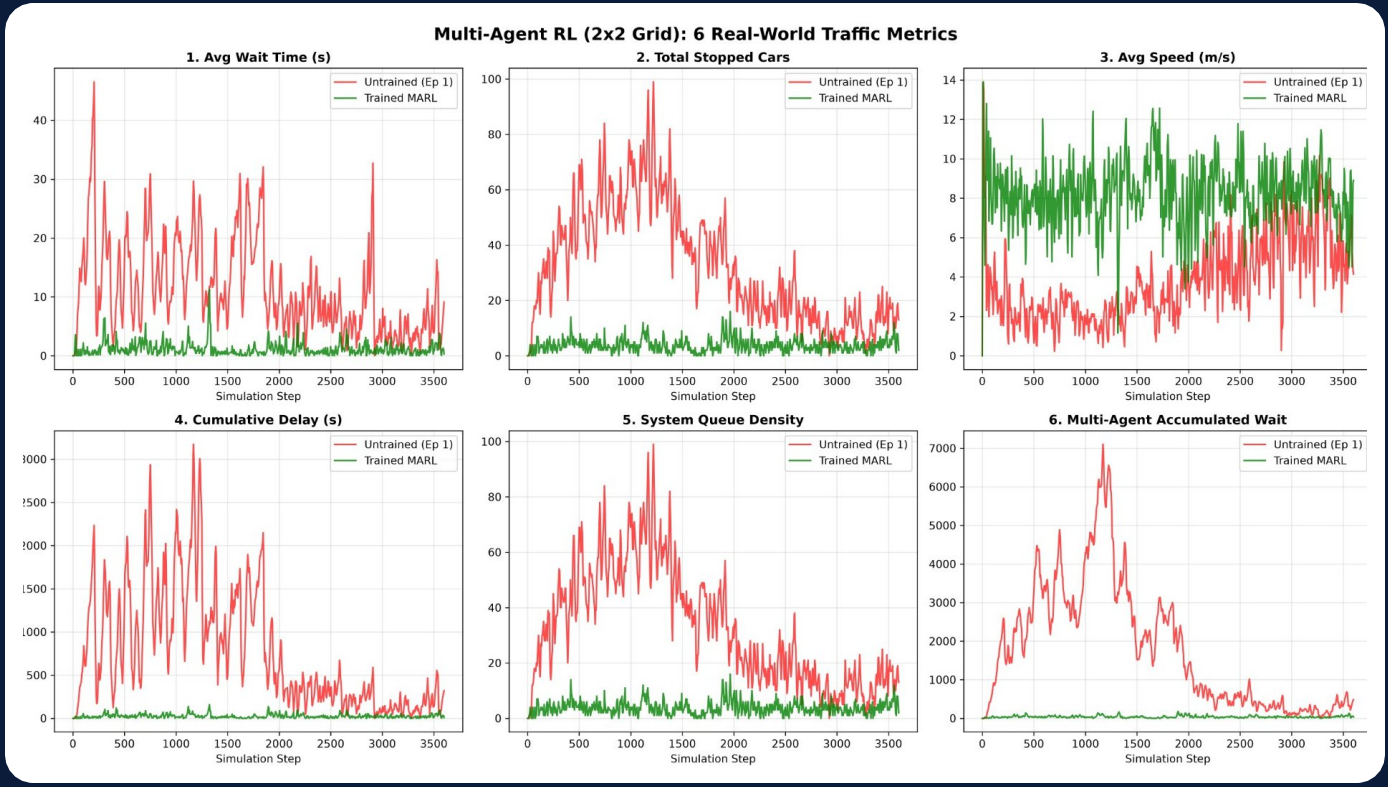
Part	Definition
State	Queue and wait per lane, current phase, time since change
Action	Next green phase, with minimum green and yellow times
Reward	Less accumulated waiting time, plus a shared network term
Algorithm	DQN with replay and a target network

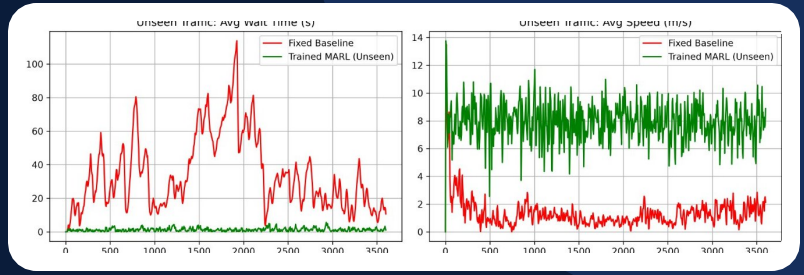


SUMO 2<2 grid · 4 agents · 3,600 steps per episode · dots mark controlled junctions

RESULTS

Trained agents beat the baselines on every metric





<6 s — Average wait on an unseen seed; fixed timing peaks near 114 s.

≈8 m/s — Average speed; fixed timing manages 1 to 2 m/s.

<200 s — Accumulated wait after training; the untrained agent peaks near 7,100 s.

Left: trained agents against episode 1. Right: unseen seed against fixed timing. Values read from plots.

LIMITATIONS

What these results do not show

Small, simulated network	Four junctions in SUMO; a real city adds irregular layouts and sensor noise.
Other layouts untested	The policy generalises across unseen seeds; other road layouts and demand patterns are future work.
Simple traffic mix	No pedestrians, buses, cyclists or emergency vehicles in the model.
Scaling and sim-to-real gap	Larger grids need more training, and a real deployment needs safe testing first.

ETHICS

Faster traffic must stay fair and safe

Fairness

Do not starve minor roads or pedestrians to speed up main flows.

Safety

Keep minimum green and yellow times, give emergency vehicles priority, and fall back to fixed timing if sensors fail.

Privacy

Use anonymous lane counts, not vehicle or licence-plate tracking.

Accountability

Log decisions, keep human oversight, and audit the reward for unintended trade-offs.

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